



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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Section: BSIT - 2B	Instructor:

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: This program is designed to serve as an IT automation Incident Ticket Manager that helps support teams organize and track bot error reports. It uses a linear data structure in Python to store incident records sequentially in memory. Through an interactive text menu, the program allows users to easily add new tickets, view all active issues, search by ticket ID, delete resolved cases, and check the total number of unresolved incidents.

Program Operations: Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

Code line: `tickets.append([input("Incident ID: "), input("Bot: "), input("Description: ")])`

Explanation: it Collects user inputs for ID, Bot name, and Description into an inner list, then uses `.append()` to attach it directly to the end of the tickets list.

Display Incidents:

Code line: `for t in tickets:`

`print(t[0], "[", t[1], "]", t[2])`

Explanation: It Loops sequentially through the outer list and prints the contents of each sublist using index positions 0(ID), 1(Bot), and 2 (Description)

Search Incident:

Code line: `for t in tickets:`

`if t[0] == id:`

`print("Found: ", t[0], "[", t[1], "]", t[2])`

`break`

Explanation: It performs a searching across the list by comparing the user input against the incident ID of each ticket until a match is found and printed.

Remove Resolved Incident:

Code line: `for i in range(len(tickets)):`

`if tickets[i][0] == id:`

`ticket.pop(i)`



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```
print("Removed")  
break
```

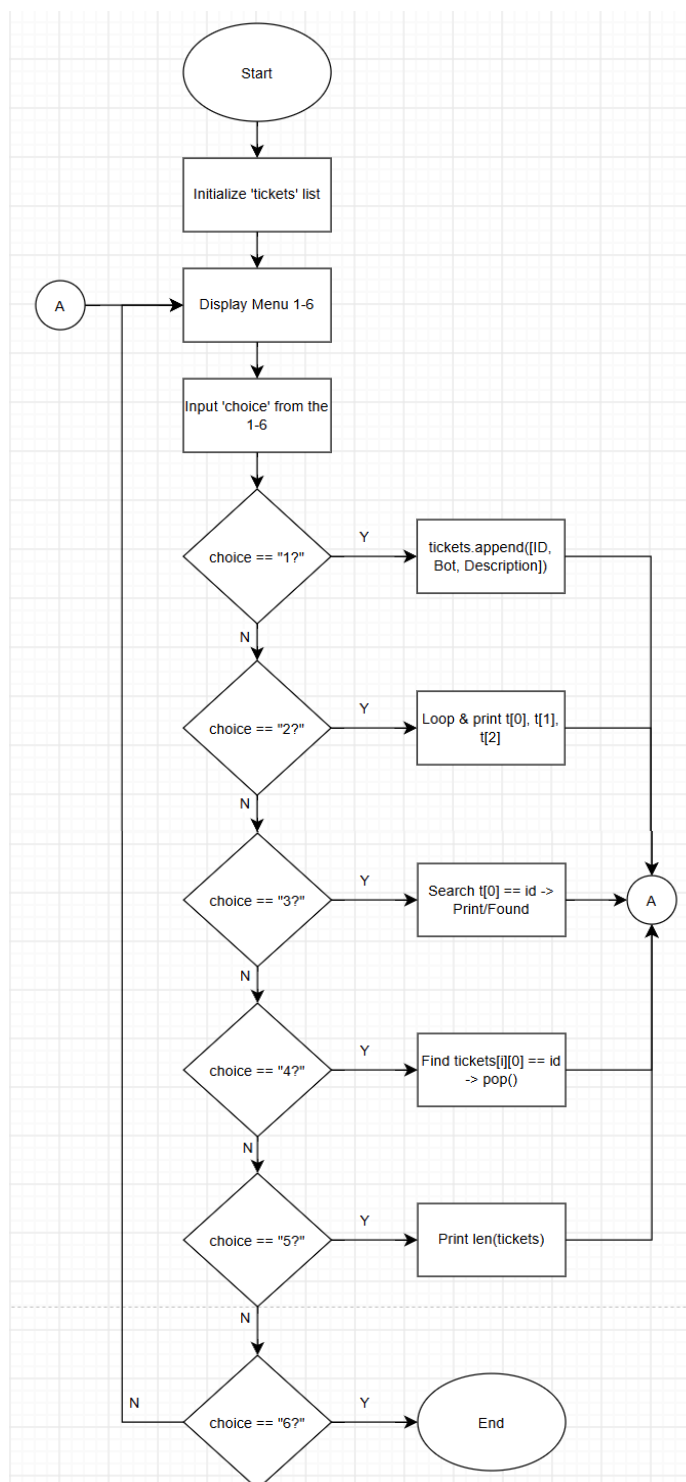
Explanation: It iterates through index positions using `range(len(tickets))`, compares `tickets[i][0]` with the ID, and deletes the matching record using `tickets.pop(i)`

Count Active Incidents:

Code line: `print("Active tickets:", len(tickets))`

Explanation: Calls Python's `len()` function to get the current number of sublists stored in `tickets` and prints the count.

Flowchart: Create a flowchart showing the overall process and logic of your program.





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- II. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

In this activity, I learned how sequential linear data structures can be applied in Python using nested lists to manage real-world IT incident logs. The linear data structure was applied by treating each ticket as a sublist inside a primary list, allowing operations like adding elements at the end, iterating through items sequentially, using linear search by index, and removing entries with `pop()`. This activity showed how fundamental list operations and index manipulation work together to create an organized data management program